

Degradation modes of large-format stationary-storage LFP-based lithium-ion cells during calendaric and cyclic aging

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Abstract

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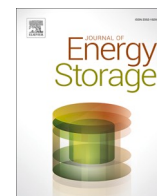
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ABSTRACT

Lithium-ion batteries exhibit capacity loss as a result of the combined degrading effects of calendaric and cyclic aging. In this study, we quantify the lifetime of large-format (180 Ah) commercial stationary-storage lithium iron phosphate-based lithium-ion cells by performing 1500 cycles of cyclic aging and ca. 850 days of calendaric aging. The aging tests were performed at two different temperatures (35 °C and 50 °C) to observe the effect of temperature on aging. The calendaric aging cells were stored at two different states of charge (SOC) (100 % and 75 %) to observe the effect of SOC. At the end of aging tests, the capacity loss of all cells at 50 °C exceeded those of all cells at 35 °C. Temperature was thus identified as major aging driver. The observed global activation energy over all investigated aging protocols was 37.3 kJ/mol. Furthermore, aging modes (loss of lithium inventory and loss of active material) were investigated by differential voltage analysis of the charge-discharge curves; for the cyclic aging cells, this was performed on the cycling data directly. The degradation mode analysis showed that loss of lithium inventory is mainly responsible for capacity loss.

1. Introduction

Harvesting solar energy by using photovoltaic (PV) panels is the most popular carbon-free residential electricity generation technology. Between 2015 and 2022, the installed global capacity of residential solar PV power increased by 400 % from 38.9 GW to 195.2 GW [1]. Electrical energy storage is an indispensable element of an energy system with a high ratio of renewable energies to compensate supply-demand mismatch. In 2023, the installed energy of residential energy storage in Germany increased by 165 % compared to the installed energy in 2022, with 98 % of the systems equipped with lithium-ion batteries [2,3]. A residential battery energy storage system (BESS) is generally consisting of multiple battery cells, an inverter to provide alternating current output, and an energy management system to control the operation and ensure system safety. Despite the continuously decreasing prices [4,5], cell costs remain the biggest expense in BESS [6]. Therefore, a long cell lifetime is vital for their economic feasibility. The investigation of cell aging has significance because understanding the influence of operating conditions (temperature, charge/discharge profile and state of charge, SOC) and aging mechanisms contributes to improving cell performance. Moreover, understanding aging is important for accurate lifetime prognosis.

Published lifetime data from commercial BESS operation are scarce. Figgenger et al. have recently published a field study on 8 years of operation of 21 privately-owned home-storage systems [7]. Most of the investigated systems reached end of life (defined as 20 % capacity loss) after 1000-1500 equivalent full cycles or 5–7 years. Dubarry et al. have evaluated operation data from a 2.3 GWh BESS used for frequency regulation [8]. They observed around 15 % capacity loss after 7 years and 7500 equivalent full cycles. We have investigated two 8 kWh home-storage systems under laboratory operation and observed end of life after around 800 equivalent full cycles (2.5 years) [9]. These studies show that aging is significant, and practical lifetimes of BESS are far lower than those of PV systems.

One of the common cell chemistries used in residential BESS is graphite/lithium iron phosphate (LiFePO₄, LFP) due to its relatively low cost, long lifetime and thermal stability [10–12]. We have recently investigated aging of 28 commercial LFP lithium-ion cells as a part of two 8 kWh commercial energy storage systems without active temperature control during 2.5 years of operation [9]. We observed significantly faster capacity loss in those systems compared to a single cell cycled under controlled laboratory conditions. We concluded that cell temperature, rather than cycle number, has a dominant effect on capacity loss [9]. Various studies on the effect of temperature on the aging

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of LFP cells demonstrated higher capacity losses at temperatures above 40 °C compared to room temperature during cyclic [13–17] or calendaric aging [18,19]. Moreover, during calendaric aging, higher capacity loss was observed at elevated temperatures compared to room temperature independent of SOC, thus confirming the dominant effect of temperature on aging [20,21].

Studies available for LFP cell aging have been in general performed on cells with a relatively small capacity (≤ 15 Ah). However, cell format, size and internal structure have a strong influence on the internal temperature distribution and, hence, aging. Therefore, results from small-format cells are not necessarily representative of the behavior of large-format cells. In the current study we investigate large-format (180 Ah nominal capacity) LFP cells for usage in residential BESS. To our knowledge this is the first time commercial large-format cells have been investigated for a long period of ca. 850 days.

For lithium-ion batteries under real operation conditions, physicochemical aging mechanisms cannot be easily identified from the limited measurement data available, which typically include only voltage, current, and temperature. In order to facilitate analysis, physicochemical aging mechanisms can be grouped into degradation modes whose effects can be quantified by changes in the open circuit voltage (OCV) curves [22]. In this study, we follow the degradation modes approach and accordingly group the aging mechanisms into two degradation modes: loss of active material (LAM) and loss of lithium inventory (LLI) [23]. Both modes nonlinearly reduce cell capacity and increase internal resistance (IR). LAM is caused by different mechanisms, such as the isolation of active material particles due to particle cracking [23,24], the formation of fully passivating surface layers [25,26], or electrode dry-out [27]. The amount of charge is limited with available lithium inventory that can participate in redox reactions. Thus, LLI causes irreversible loss of cell capacity. The most dominant mechanism causing LLI is the formation of the solid electrolyte interphase (SEI) layer due to the decomposition of the electrolyte [10,18,19,28–32]. Following Dubarry's degradation mode modeling approach [33], the full-cell voltage response is a convolution of the half-cell voltage responses, depending on the two electrodes' relative capacities ("loading ratio") and a shift of the half-cell curves against each other ("offset"). From a physically-based viewpoint, these properties are directly related to the lithium stoichiometry ranges in the active materials during cycling. They are also referred to as the *balance* of the electrodes [34]. Time-resolved analysis of the electrode balance allows to quantify LAM and LLI. This concept has been widely adopted in the lithium-ion battery aging literature [22]. In this study, we apply it to the calendaric and cyclic aging data of the investigated large-format cells.

The goal of the present paper is to investigate the aging behavior of large-format LFP-based lithium-ion cells by quantifying electrical quantities (capacity, IR, efficiency), electrode balances (lithium stoichiometry limits) and degradation modes (LAM, LLI) as function of time for different aging conditions (cyclic vs. calendaric) and temperature. The experimental methodologies and degradation mode analysis are described in Section 2. The calendaric and cyclic aging results are presented in Section 3. The results of degradation mode analysis are given in Section 4. In Section 5, the combined results are discussed. Finally, the conclusions of the study are outlined in Section 6.

2. Methodology

2.1. Investigated cells

In this study we investigated twelve individual CALB CA180FI (China Aviation Lithium Battery Co. Ltd., China) lithium-ion battery cells, as selected from the same delivery batch. The cells feature LFP as positive active material and graphite as negative active material. The cell specifications are given in Table 1. The electrical and geometrical characteristics of the cells were analyzed in detail in our previous work [35].

Table 1

Specifications of the studied cell according to the manufacturer data sheet [36] and our measurements [35].

Manufacturer	China Aviation Lithium Battery Co. Ltd. (CALB)
Model	CA180FI
Active material	Negative electrode: graphite / Positive electrode: LFP
Nominal capacity / Ah	180
Nominal voltage / V	3.2
Voltage range / V	2.5–3.65
Nominal weight / kg	5.7
Nominal energy / Wh	576
Size / mm ³	280 × 180 × 71

Each fresh cell underwent an activation procedure performed with a Biologic VMP3 battery cycler (Bio-Logic Science Instruments SAS, France) in a CTS 40/200 LI climate chamber (CTS Clima Temperatur Systeme GmbH, Germany) set to 20 °C before start of the aging experiments, with the intent of reducing effects of electrochemical (e.g., diffusion of ions to inactive sites in anode overhang) fatigue from possible long periods of inactivity during or before shipment. Activation was performed using two full constant current constant voltage (CCCV) discharge-charge cycles consisting of 50 A (0.28C) constant current (CC) phase followed by constant voltage (CV) phase with 9 A (0.05C) cut-off current, followed by a final full discharge; the CV phase was included for both charge and discharge. The cells rested 30 min between charge and discharge cycles.

2.2. Calendaric and cyclic aging experiments

Calendaric and cyclic cell aging was investigated with twelve individual cells. The tests were carried out using two separate climate chambers Binder KB 115 climate chambers (Binder GmbH, Germany) set to 35 °C and 50 °C ambient temperature, respectively. In each chamber, three tests, each consisting of two cells under the same conditions, were performed. A summary of the test cells and protocols is given in Table 2.

Two SOC levels were selected for calendaric aging tests: 100 % represents a fully-charged cell and 75 % represents a partially discharged cell. These two SOC levels were selected in order to investigate the sensitivity of the LFP electrode potential on calendaric aging. For both SOC levels, the graphite electrode potential is the same, being on one of the graphite potential plateaus. However, the LFP electrode shows a large potential difference of about 0.35 V between 75 % and 100 % SOC (of a fresh cell). This is shown in Fig. 1.

Calendaric aging test cells, following the initial reference performance test (RPT) (cf. Section 2.3), were charged with 10 A (0.055C) with Basytec GSM battery cycler (Basytec GmbH, Germany) until the test voltage level corresponding to the SOC level indicated in Table 2. The test voltages were calculated from the voltage values corresponding to respective SOC levels at pseudo-OCV curves (cf. Section 2.3) measured during the first RPT. Then, the cells were kept at CV for 5 h at test voltage. After the first 5 h CV phase, the calendaric aging cells were kept actively at the target SOC ("floated") by applying continuous CV at test voltages throughout calendaric aging tests. Thus, calendaric aging cells were continuously connected to the battery cycler to stay constantly at the set voltage. The influence of voltage floating on calendaric aging is typically supposed to be minor [54]. Fig. 2 shows exemplary current and voltage time profiles of a calendaric aging cell. Continuous but small current fluctuations (ca. 0.75 A, corresponding to around C/250) around an offset of ca. −0.4 A are visible, which is likely within the measurement accuracy of the battery cycler.

Cyclic aging cells underwent 1500 full CCCV cycles performed with Basytec XCTS battery cycler (Basytec GmbH, Germany) consisting of a 50 A (0.28C) CC phase followed by a CV phase with 9 A (0.05C) cut-off current; the CV phase was included for both charge and discharge. The cells rested only 10 s after each charge and discharge. The test plan was programmed to run for 100 cycles and restart to segment the test data for

Table 2
Calendaric and cyclic aging test protocols.

Test group	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6
Cell #	1, 2	3, 4	5, 6	7, 8	9, 10	11, 12
Temperature	35 °C	35 °C	35 °C	50 °C	50 °C	50 °C
Aging type	Cyclic	Calendaric	Calendaric	Cyclic	Calendaric	Calendaric
C-rate (current)	0.28C (50 A)	–	–	0.28C (50 A)	–	–
SOC (voltage)	–	75 % (3.336 V)	100 % (3.65 V)	–	75 % (3.336 V)	100 % (3.65 V)
Cycling depth	100 %	–	–	100 %	–	–

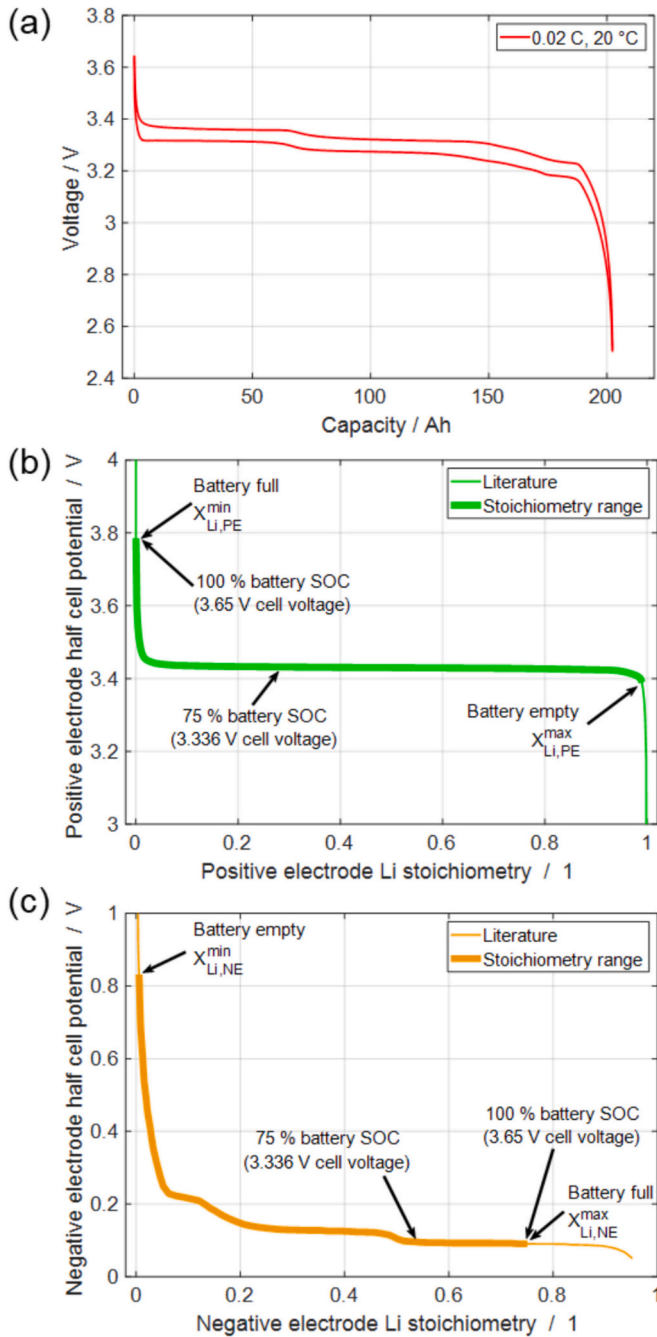


Fig. 1. (a) Low C-rate (0.02C) charge and discharge curves of a fresh Calb cell used to obtain pseudo-OCV (pOCV) curve according to the protocol in Section 2.3, (b) positive electrode, and (c) negative electrode half-cell potential curves and respective Li stoichiometry ranges used within the nominal voltage ranges, as obtained from electrode balancing analysis. The SOC selected for calendaric aging are marked. The half-cell curves were taken from literature [41,43].

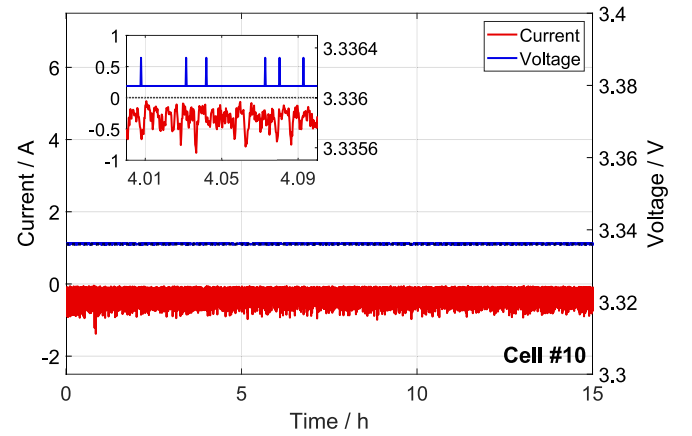


Fig. 2. Current and voltage of cell #10 (cf. Table 2) as function of time for an exemplary 15 h duration during calendaric aging at 50 °C. Black dotted line represents battery cyclers set voltage. Inset: Detail of the figure showing current and voltage fluctuations in a 6-minute timeframe.

efficient data recording and continuous data analysis.

The cycling data were analyzed for each individual cycle. The actual cell capacity (C_{cell}) was calculated by integrating the current over the discharge duration. Internal resistance (IR_{cycle}) was calculated according to [37]

$$IR_{\text{cycle}} = \frac{\bar{V}_{\text{chg}} - \bar{V}_{\text{dis}}}{\bar{I}_{\text{chg}} - \bar{I}_{\text{dis}}} \quad (1)$$

where $\bar{V}_{\text{chg}}$ is the average charge voltage, $\bar{V}_{\text{dis}}$ is the average discharge voltage, $\bar{I}_{\text{chg}}$ is the average charge current, and $\bar{I}_{\text{dis}}$ is the average discharge current in full charge/discharge cycle (the sign of charge current is positive and the sign of discharge current is negative). Capacity-based state of health (SOH_c) was calculated as

$$SOH_c = \frac{C}{C_0} \quad (2)$$

where C_0 is the initial cell capacity calculated during the initial RPT. Coulombic efficiency (CE) and energy efficiency (η_{energy}) were calculated as

$$CE = \frac{Q_{\text{discharge,CCCV}}}{Q_{\text{charge,CCCV}}} \quad (3)$$

$$\eta_{\text{energy}} = \frac{E_{\text{discharge,CCCV}}}{E_{\text{charge,CCCV}}} = \frac{\int_{\text{discharge}} I(t) \cdot V(t) \cdot dt}{\int_{\text{charge}} I(t) \cdot V(t) \cdot dt} \quad (4)$$

where $Q_{\text{discharge,CCCV}}$ is charge throughput during discharge, $Q_{\text{charge,CCCV}}$ is charge throughput during charge, $E_{\text{discharge,CCCV}}$ is the electrical energy supplied from the cell during discharge and $E_{\text{charge,CCCV}}$ is the electrical energy supplied to the cell during charge.

2.3. Periodic reference performance tests (checkups)

The status of all cells was monitored by four periodic reference performance tests (RPT) (hereafter also referred to as “checkup”) performed when cyclic aging cells made 100, 500, 1000, and 1500 full cycles. The checkups of the calendaric aging cells were performed simultaneously with the cyclic aging cells. The checkup intervals correspond to 106, 255, 553, and 846 days of calendaric aging (slightly differing for the individual cells). The objective of the first checkup after start of the tests (checkup #1), performed at 100 cycles (106 days) of aging, was to test the setup and not to interrupt aging tests for long period of time. Therefore, during first checkup time-consuming low C-rate pseudo-OCV (pOCV) tests (see below) were not performed. Thus, these were not used for degradation modes analysis. The initial RPT (checkup #0) was performed with a Biologic VMP3 battery cycler inside CTS-40/200 LI climate chambers. The following RPTs (checkups #1-4) were performed with Basytec XTS and Basytec GSM battery cyclers inside Binder KB 115 climate chambers. Before each checkup, the cells were kept for at least one day at 20 °C to ensure thermal equilibrium. The initial and following checkups were performed at 20 °C ambient temperature following the same protocol consisting of capacity, IR, and pOCV measurements.

First step of the checkup was the determination of actual cell capacity. Capacity tests consisted of two full charge/discharge cycles identical to the activation test protocol (cf. Section 2.1). The discharge capacity of the second cycle was defined as actual cell capacity, C_{cell} , obtained by integrating the measured current over the complete discharge duration (including CC and CV phases). The capacity measured during the initial checkup was defined as the initial cell capacity, C_0 . The accuracy of the capacity measurement, based on the current measurement accuracy of the cyclers as given by the cycler manufacturers, is estimated around 1.6 Ah, which corresponds to 0.8 % of the initial capacity. This also limits the accuracy of CE and η_{energy} to 0.8 %.

In a second step, IR measurements were performed at each 2 % SOC from 98 % SOC to 2 % SOC, based on C_{cell} . IR was measured by applying a lower 25 A (0.14C) current pulse for 30 s, starting from t_0 , during 50 A (0.28C) continuous CC discharge or charge. From the voltage response, IR was calculated according to

$$\text{IR}_{\text{pulse}} = -\frac{V(t_1) - V(t_0)}{I(t_1) - I(t_0)} \quad (5)$$

by taking $t_1 - t_0 = 3$ s. For cell selection (cf. Section 3.1), the average IR in the whole SOC range calculated from Eq. (5) (IR_{pulse}) during the initial RPT (checkup #0) is used to compare homogeneity of cells. For aging investigations, IR calculated from charge/discharge curves during checkups according to Eq. (1) (IR_{cycle}) is used for the sake of accurate comparison with the IR calculated from aging cycles by using the same approach.

Finally, one full CCCV charge-discharge cycle at low-C-rate (0.1C = 18 A, 9 A CV phase cut-off current, 30 min rest between charge-discharge) was performed at 20 °C in order to calculate OCV curves. There is a limitation to the accuracy of the battery cyclers, so reducing the C-rate further would decrease the signal-to-noise ratio considerably and increase checkup time significantly. Furthermore, LFP-based cells exhibit a well-known voltage hysteresis, that is, the OCV is different depending on whether the cell is being charged or discharged [38–40]. Therefore, we took the arithmetic average of charge and discharge voltage corresponding to the same SOC. Thus, we interpret the resulting characteristic curve as pseudo-OCV curve.

The RPTs during aging experiments (checkups #1-4) were performed without changing the cell position in order to avoid potential influences from a change in cooling profiles as a result of cell displacement inside the climate chambers.

2.4. Analysis of electrode balance

The “balance” of the positive electrode (PE) and negative electrode (NE) refers to the range of lithium stoichiometries (x in Li_xC_6 for graphite, y in Li_yFePO_4 for LFP) that the cell uses during cycling between the cut-off voltages. We define the lithium stoichiometries x and y as the lithium concentration relative to the maximum lithium concentration according to

$$x \text{ or } y = \frac{c(\text{Li})}{c_{\text{max}}(\text{Li})} \quad (6)$$

where $c(\text{Li})$ is the actual lithium concentration and $c_{\text{max}}(\text{Li})$ is the maximum lithium concentration in graphite (NE) or LFP (PE). Theoretically, lithium stoichiometry varies between 0 and 1. However, in practice, the complete range may not be accessible.

The aim of electrode balance analysis is to empirically determine the actual stoichiometry ranges of both electrodes during a full cycle using pOCV measurements, which is needed to further quantify LLI am LAM. The methodology to determine the electrode balance from measured pOCV curves was developed in-house and presented in Ref. [34]. Briefly, the full-cell pOCV curve $V_{\text{cell}}^{\text{eq}}$, depending on the full-cell charge throughput Q , is a superposition of the two half-cell potential curves, $V_{\text{PE}}^{\text{eq}}$ (positive electrode) and $V_{\text{NE}}^{\text{eq}}$ (negative electrode), depending on the half-cell lithium stoichiometries X_{PE} (positive electrode), X_{NE} (negative electrode). The stoichiometry ranges were determined by comparing full-cell OCV curves synthesized from the half-cell curves to the measured pOCV of the target cell using mathematical optimization. Mathematically, this is described as the nonlinear optimization problem

$$\min_p \chi^2 = \min_p \left[\sum_{n=1}^N \left(\frac{d}{dQ} (V_{\text{cell,exp}}^{\text{eq}})_n - \frac{d}{dQ} (V_{\text{cell,mod}}^{\text{eq}})_n \right)^2 \right] \quad (7)$$

with five unknown parameters,

$$p = \{X_{\text{PE}}^{\text{min}}, X_{\text{PE}}^{\text{max}}, X_{\text{NE}}^{\text{min}}, X_{\text{NE}}^{\text{max}}, C_{\text{cell}}\} \quad (8)$$

Here, $V_{\text{cell,exp}}^{\text{eq}}$ is the measured pOCV curve, $V_{\text{cell,mod}}^{\text{eq}}$ is the synthetic OCV curve calculated from the half-cell potential curves, χ^2 is the merit function, and the sum runs over N measured data points of the pOCV curve. The merit function represents the deviation between experimental and synthetic differential voltage (DV) curves. The required methods for interpolation, table manipulation, and optimization were taken from MATLAB (R2019a). The interior-point algorithm implemented in MATLAB's *fmincon* solver was used for optimization. In order to create a synthetic OCV curve ($V_{\text{cell,mod}}^{\text{eq}}$), the half-cell potential curves of the two active materials (AM) are needed. As half-cell measurements were not included in our experimental framework, we used literature data. Half-cell potential curves of a given active material may vary depending on the source and processing (e.g., degree of graphitization of graphite). We tested different data for LFP and for graphite, and obtained the best optimization results with the following data. For LFP, half-cell potential curves were taken from Verbrugge et al. [41], and entropies from Viswanathan et al. [42]. For graphite, half-cell potential curves were taken from Smekens et al. [43], and entropies from Reynier et al. [44].

Fig. 1 shows exemplary results of electrode balance analysis of an as-received cell. Half-cell curves from the literature are shown as thin solid lines in panels (b) and (c), whereas the calculated stoichiometry ranges are represented as thick solid lines. The end points of the used stoichiometry ranges represent the full and empty battery states.

Electrode balance analysis was applied to both, the pOCV data obtained during checkups, and the continuous cycles of the cyclic aging test cells. When analyzing the cyclic aging test cells, the last optimized parameters were used as initial values for the next optimization.

2.5. Degradation modes

From the electrode balance obtained in the previous section, three degradation modes can be obtained, that is, loss of active material of the positive electrode (LAM_{PE}), loss of active material of the negative electrode (LAM_{NE}), and loss of lithium inventory (LLI). We quantify them in units of capacity Q (in Ah) for straightforward interpretation.

The stoichiometry range ΔX that an electrode is cycled in between the cell cut-off voltages is obtained from the upper and lower stoichiometry limits ($X^{\max}$ and $X^{\min}$) calculated from electrode balance analysis as

$$\Delta X(t) = X^{\max}(t) - X^{\min}(t) \quad (9)$$

for both electrodes. The theoretically available half-cell capacities for positive and negative electrodes (C_{PE} and C_{NE}) can be obtained as

$$C_{PE}(t) = \frac{C_{cell}(t)}{\Delta X_{PE}(t)} \quad (10)$$

$$C_{NE}(t) = \frac{C_{cell}(t)}{\Delta X_{NE}(t)} \quad (11)$$

Eqs. (9)–(11) are time-dependent because stoichiometric ranges and cell capacity change with aging. For a fresh cell, the equations above are redefined by adding the superscript “0” to indicate that all values address the initial values as

$$C_{PE}^0 = \frac{C_{cell}^0}{\Delta X_{PE}^0} \quad (12)$$

$$C_{NE}^0 = \frac{C_{cell}^0}{\Delta X_{NE}^0} \quad (13)$$

We assume that the fresh cell has no LAM. LAM_{PE} and LAM_{NE} of the aged cell can therefore be calculated as

$$Q_{LAM,PE}(t) = C_{PE}^0 - C_{PE}(t) \quad (14)$$

$$Q_{LAM,NE}(t) = C_{NE}^0 - C_{NE}(t) \quad (15)$$

LLI causes a shift of the two half-cell curves with respect to each other. In order to develop a quantitative description, the charge content of the two electrodes Q_{PE} , Q_{NE} can be defined according to

$$Q_{NE}(t) = X_{NE}(t)C_{NE}(t) \quad (16)$$

$$Q_{PE}(t) = X_{PE}(t)C_{PE}(t) \quad (17)$$

We assume that the cell is manufactured with a fully-lithiated positive electrode ($X_{PE}^{\max} = 1$, $Q_{PE} = C_{PE}^0$, here: LiFePO₄) and a fully delithiated negative electrode ($X_{NE}^{\min} = 0$, $Q_{NE} = 0$, here: pure graphite C₆). We furthermore assume that LLI occurs only at the NE, resulting from SEI formation and irreversible plating. With these assumptions, the NE lithium stoichiometry at an arbitrary time is given as

$$X_{NE}(t) = \frac{Q_{NE}(t)}{C_{NE}(t)} = \frac{Q_{cell}(t) - Q_{LLI}(t)}{C_{NE}(t)} \quad (18)$$

and the PE lithium stoichiometry is given as

$$X_{PE}(t) = \frac{Q_{PE}(t)}{C_{PE}(t)} = \frac{C_{PE}^0 - Q_{cell}(t)}{C_{PE}(t)} \quad (19)$$

Here, $Q_{cell}(t)$ is the total charge throughput of the cell until time t . For a fully charged cell $Q_{cell} = C_{cell}$ and for a fully discharged cell $Q_{cell} = 0$. The combination of Eqs. (18) and (19) by eliminating $Q_{cell}(t)$ results in an expression for Q_{LLI} according to

$$Q_{LLI}(t) = C_{PE}^0 - X_{PE}(t)C_{PE}(t) - X_{NE}(t)C_{NE}(t) \quad (20)$$

The expression can be calculated with an arbitrary pair of $X_{PE}(t)$,

$X_{NE}(t)$. Due to the better identifiability of X_{PE} for a fully-charged cell (cf. next section), we used the stoichiometric pair at 100 % SOC, resulting in

$$Q_{LLI}(t) = C_{PE}^0 - X_{PE}^{\min}(t)C_{PE}(t) - X_{NE}^{\max}(t)C_{NE}(t) \quad (21)$$

Eqs. (14), (15) and (21) allow to quantify LAM_{PE}, LAM_{NE}, and LLI from electrode balance analysis of a fresh cell (results indicated with superscript ⁰) and the same cell after aging for the duration of time t .

2.6. Parameter identifiability

The framework derived in the two previous sections allows the determination of LAM_{PE}, LAM_{NE}, and LLI by using parameters $X_{PE}^{\min}$, $X_{PE}^{\max}$, $X_{NE}^{\min}$, $X_{NE}^{\max}$, and C_{cell} . From the experimental data, however, not all of these parameters are identifiable. Therefore, further assumptions are necessary to eliminate the unidentifiable parameters. In Fig. 1 it is noticeable that with a fully-charged cell the NE lithium stoichiometry is in a potential plateau region, while the PE half-cell curve of the cathode shows a strong gradient. For an empty cell, on the other hand, the NE half-cell curve exhibits a strong gradient and the PE is adjacent to the voltage plateau, spanning nearly the whole PE stoichiometry. Therefore, the identifiability of X_{NE} is best for an empty cell and that of X_{PE} for a full cell. Still, as the NE half-cell potential curve has several steps, the identifiability of the NE lithium stoichiometry range remains good.

As a result of ongoing LLI, the maximum stoichiometry of the positive electrode $X_{PE}^{\max}$ shifts to lower values in time. For the LFP electrode of the cell shown in Fig. 1 (b), $X_{PE}^{\max}$ shifts into the large voltage plateau where the voltage is nearly constant. Within this plateau, $X_{PE}^{\max}$ is practically unidentifiable because the plateau voltage can be associated with a wide range of Li stoichiometries. Thus, $X_{PE}^{\max}$ needs to be excluded from the optimization problem. This, in turn, makes the identification of LAM_{PE} impossible. Therefore, we assumed that the LFP positive electrode does not undergo LAM by setting

$$Q_{LAM,PE} = 0. \quad (22)$$

Accordingly, we assume that the theoretical capacity of PE does not change,

$$C_{PE} = C_{PE}^0. \quad (23)$$

Under this assumption, the maximum stoichiometry at any given time $X_{PE}^{\max}(t)$ can be calculated according to

$$X_{PE}^{\max}(t) = X_{PE}^{\min}(t) + \frac{C_{cell}(t)}{C_{PE}^0} \quad (24)$$

It might be mentioned that LAM_{PE} is non-identifiable only as long as LAM_{PE} < LLI, which is the case in the present experiments. The quantification of LAM_{PE} would require dedicated half-cell experiments with PEs harvested from the full cell at different aging states. However, this is out of scope of the present study. If only LAM_{PE} occurs, or if LAM_{PE} > LLI, the LFP electrode cycles beyond its plateau, and the stoichiometric range remains identifiable from full-cell measurements.

Furthermore, as mentioned earlier, we calculated C_{cell} directly from the experimental data from the discharge curves. The number of optimization parameters was reduced by two with these two assumptions. As a consequence, the parameter set to be determined for solving the optimization problem was reduced to $p = \{X_{NE}^{\min}, X_{NE}^{\max}, X_{PE}^{\min}\}$. These parameters are sufficient to calculate LAM_{NE} and LLI from Eqs. (15) and (21), respectively.

3. Experimental results

3.1. Initial characterization

In order to minimize cell-to-cell variations, the most homogenous twelve cells were selected according to the capacity (C_{cell}) and IR (IR_{pulse}) measurement results during initial RPT (checkup #0), which

are shown in Fig. 3. The cells were numbered from #1 to #12 for aging tests (cf. Table 2). The average discharge capacities vary between 199.9 Ah and 201.1 Ah. The capacity range is 1.2 Ah which corresponds to 0.6 % of the maximum cell capacity. The average (between 2 and 98 % SOC) IR varies between 0.58 m Ω and 0.73 m Ω . The IR range is 0.15 m Ω , corresponding to 20 % of maximum IR. While capacity variation is small, IR variation is significantly higher than reported in literature [55]. This may be explained by the very low, sub-milliohm IR of these large-format cells that could significantly be affected by different contact resistances as a result of using different test connectors for each cell. However, the influence of contact resistance on measured IR was not further investigated.

3.2. Cyclic aging

The cyclic aging tests were performed until each cell completed approximately 1500 full cycles. Specifically, 1509, 1500, 1495 and 1505 full cycles were completed for cells #1, #2, #7 and #8, respectively. Exemplary full charge/discharge cycles of cell #1 (cycling at 35 °C) and cell #7 (cycling at 50 °C) are shown in Fig. 4. The upper panels of Fig. 4 show relatively fresh cells undergoing 10th and 11th cycles. The length of full charge/discharge cycles are 8 h (± 4 min). The lower panels show the same cells in their aged state undergoing 1100th and 1101st cycles. For the aged cells, full cycle duration is significantly shortened to 7.35 h (cell #1 at 35 °C) and 6.7 h (cell #7 at 50 °C) due to capacity loss with aging (current level was kept unchanged). Cell surface temperatures are, on average, 2 °C higher than ambient temperature (note the accuracy of the temperature sensors was verified to be ≤ 0.3 °C). CV phases were carried out for both charge and discharge, but are not visible in the data due to their very short duration (maximum 2 min) compared to the total length of full charge/discharge.

Charge and discharge curves of every 100th cycle, plotted as voltage vs. charge throughput, are shown in Fig. 5 for cell #1 (cycling at 35 °C) and cell #7 (cycling at 50 °C). The cell capacity clearly decreases with increasing number of cycles. As expected, the capacity loss is faster at 50 °C. Interestingly, at 35 °C, there is no significant change in the voltage gap between charge and discharge branches with increasing cycle number, which means no significant change in overpotentials. These stable charge-discharge plateaus indicate a non-increasing cell resistance [31]. On the other hand, at 50 °C, overpotentials increase continuously, more significantly at lower SOC (0–25 %) and higher SOC (75–100 %). Both datasets show a clear shift of the characteristic voltage step at around 60 Ah charge throughput, associated with the graphite negative electrode, which will be exploited in the degradation modes

analysis (cf. Section 4).

The quantitative analysis of cell characteristics during cycling is shown in Fig. 6. Cell #1 and cell #2 were cycled at 35 °C, and cell #7 and cell #8 were cycled at 50 °C ambient temperature. Discharge capacity is displayed in panel (a). All cells continuously lose capacity. The curves show small discontinuities every ca. 100 cycles, associated with scheduled stops and restarts of the cycling protocol according to the experimental methodology (cf. Section 2.2). The 50 °C cells show more significant discontinuities at 100, 500, 1000 and 1400 cycles. These points correspond to long interruptions of cyclic aging tests lasting more than a week: the first three are RPT and the last one corresponds to a technical problem resulting in power outage in the laboratory. After each break, the cycling continues with visibly different capacity for a few cycles until the curve approaches the pre-check-up values. This behavior has been interpreted before due to anode overhang [45]. Although the 35 °C cells were subjected to checkups simultaneously with 50 °C cells, the impact of breaks on the discharge capacity of 35 °C cells are much less pronounced than in the case of the 50 °C cells.

Fig. 6(b) shows IR_{cyclic} (cf. Eq. (1)) as function of cycle number. All cells show a small drop of IR during the first ca. 20 cycles. The origin of the temporary increase in IR between around 50–100 cycles is unclear. The 35 °C cells then exhibit a very slow increase of only 4.7 % in average between 100th and 1000th cycles, followed by a slightly accelerated IR increase. The 50 °C cells start with a significantly lower IR, associated with faster electrochemical reactions and transport processes at higher temperatures. The IR is only slightly increasing up to ca. 800 cycles. Then the cells show an “elbow” type behavior [46], with rapidly increasing IR, accompanied by short-term fluctuations. After the break at 1000th cycle for checkup #3, the cells continue with a higher IR; while after the break at 1400th cycle due to power outage, IR decreases significantly. These results underline the influence of the checkups and other long interruptions on the cell behavior, and therefore the nonlinear dependence of aging on cell history.

The IR measurement results are consistent with initial overvoltage decrease of LFP cells observed in literature [39]. The impedance decrease in first cycles of LFP cells was attributed to the electrochemical milling of the PE [47]. Electrochemical milling is the process in which LFP particles crack and crumble because of mechanical stress as a result of volumetric change during Li (de)intercalation. These cracks increase active surface area and the tortuosity of the electrode that in turn facilitates electronic conductivity and mass transport. Although beneficial in the primary phases of cycling, the detachment of such particles as a result of further milling causes detachments from conductive carbon black causes LAM. If disconnected LFP particles are lithiated, this lithium inventory is also lost [23].

Panels (c) and (d) show the state of health (SOH_C), defined here as relative capacity according to Eq. (2). Panel (c) shows a linear plot, while panel (d) uses the same data plotted as function of the square root of the number of cycles. The cells cycled at 35 °C show a significantly better capacity retention (86–88 % SOH_C) at the end of the test, compared to the ones cycled at 50 °C (77–79 % SOH_C). The cells cycled at 50 °C (cell #7 and cell #8) reached 80 % SOH_C (end of commercial life), at around 1350 cycles. The dashed line in panel (d) corresponds to linear fitting using the first 300 cycles for each cell. Cell #1 (cycled at 35 °C) demonstrates close-to-linear behavior of SOH_C versus square root of cycle number during first 500 cycles, while cell #2 (cycled at 35 °C) temporarily deviates from linear fit between cycles 50–290. After 415 cycles, cell #2 permanently deviates from linear fit. Both cells cycled at 50 °C show an initial deviation from linear fit (between cycles 27–255 for cell #7 and cycles 27–132 for cell #8). After ca. 350 cycles both cells cycled at 50 °C deviate from the linear fit permanently. In the literature, the square root of time dependence of capacity loss was experimentally validated by semi-empirical life models [48]. The physical origin of the $t^{0.5}$ dependency is attributed to diffusion-limited growth of the SEI [49].

Finally, panels (e) and (f) show CE and η_{energy} . CE remains close to

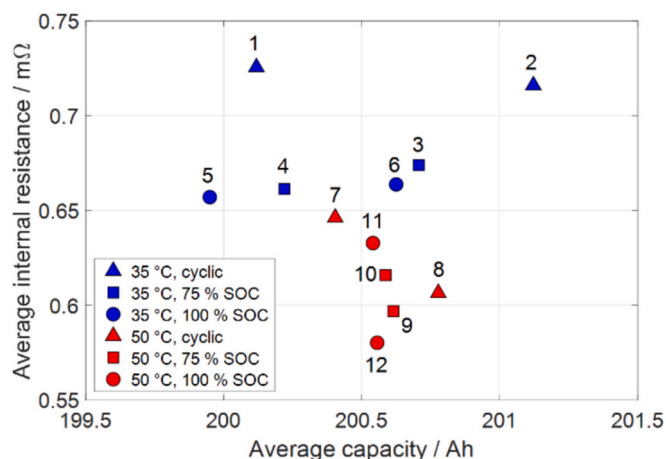


Fig. 3. Discharge capacity and average (between 2 % and 98 % SOC) internal resistance IR_{pulse} (Eq. (5)) of the twelve cells used for aging tests during initial checkup #0 (initial RPT). Cell numbers are given next to the data points (cf. Table 2). Aging test protocols of the cells are indicated in the legend.

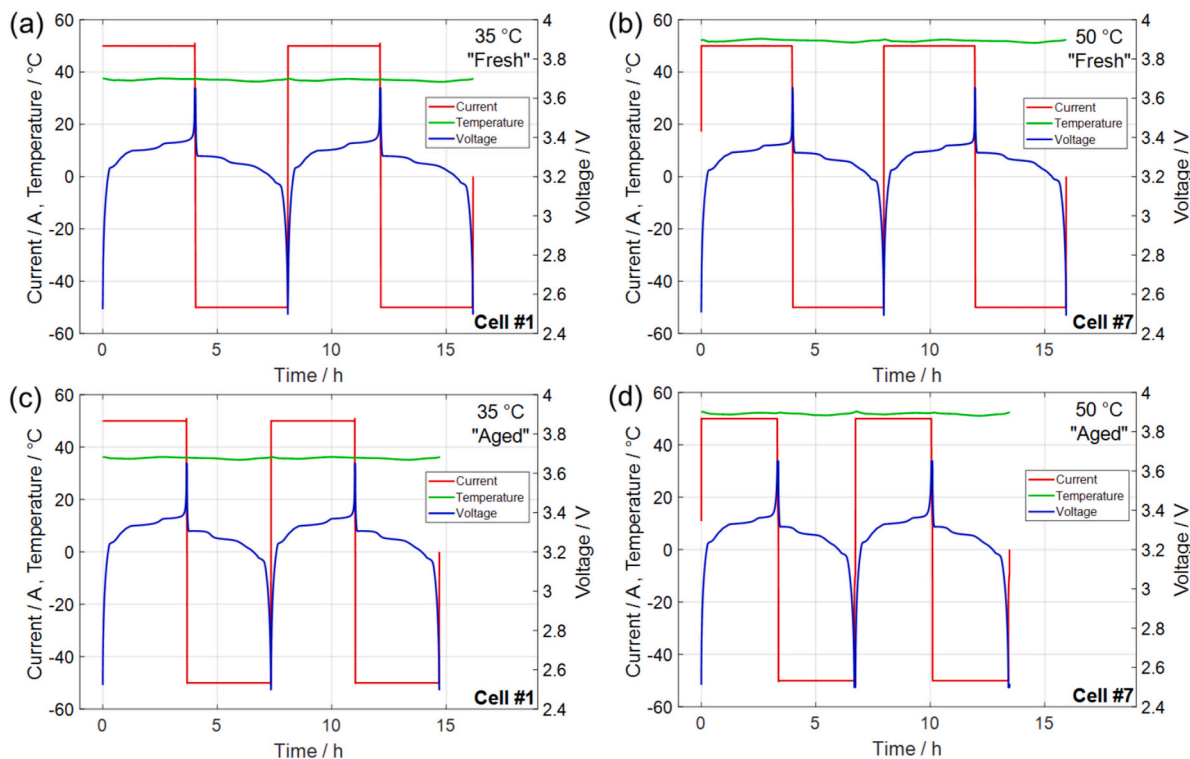


Fig. 4. Voltage, current and temperature profiles during two exemplary consecutive full cycles as function of time. (a) Cycle 10 and 11, cell #1 cycled at 35 °C. (b) Cycle 10 and 11, cell #7 cycled at 50 °C. (c) Cycle 1100 and 1101, cell #1 cycled at 35 °C. (d) Cycle 1100 and 1101, cell #7 cycled at 50 °C.

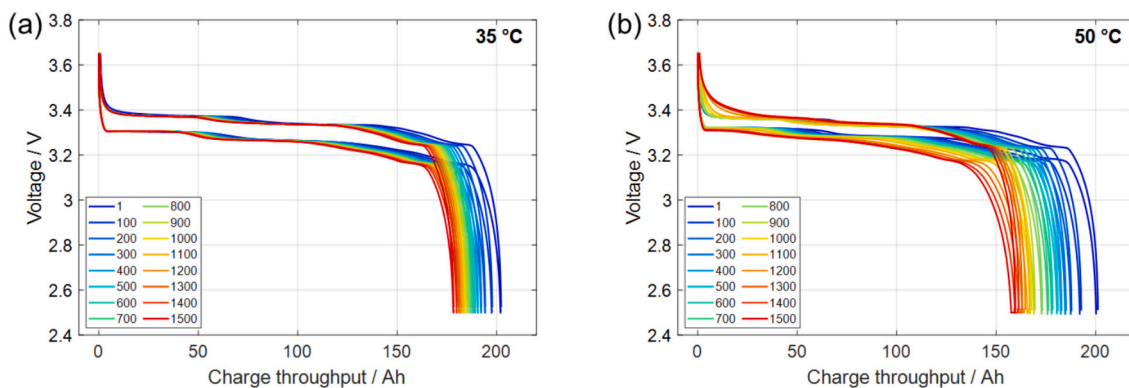


Fig. 5. Selected 0.28C aging cycles of (a) cell #1 cycled at 35 °C, and (b) cell #7 cycled at 50 °C. The lower branches on the curves represent discharge, where time progresses from left to right; the upper branches of the curves represent charge, where time progresses from right to left.

unity for all cells over the complete experiment; we believe that slightly lower efficiency of cell #7 as well as efficiencies $>100\%$ for cell #1 are due to miscalibrations of the associated battery cycler test channels; the estimated uncertainty of the CE calculation is $\pm 0.8\%$ (cf. Section 2.3). Still, the 50 °C cell data exhibit larger noise than the 35 °C cell data, the origin of which is unclear. After ca. 900 cycles, η_{energy} of 50 °C cells reduce below the efficiencies of 35 °C cells. Moreover, the reduction of η_{energy} of 50 °C cells are inversely symmetrical with the increase of IR after 900 cycles in panel (b). Therefore, IR increase is the main cause of η_{energy} decrease in 50 °C cells.

3.3. Calendaric aging

The calendaric aging cells were kept at test conditions given in Table 2 for 875 days. The quantitative analysis of checkup results is presented in Fig. 7. Checkups were performed at ca. 0, 100, 250, 550 and

850 days. Worthwhile noting that the electrical properties of the calendaric aging cells, except cell voltage, were only measured during checkups, which were carried out at 20 °C ambient temperature; therefore, these data need to be interpreted differently to those of the cyclic aging cells presented in the previous section.

Fig. 7(a) shows the discharge capacity. All cells lose capacity continuously with progressing calendaric aging. Exceptionally, between checkup #1 and checkup #2, taking place after 106 and 255 days of aging, capacity losses of 35 °C cells are minimal (between 0.8 Ah and 1.08 Ah) which is more than 2 to 7 times smaller compared to capacity losses between other checkups. SOH_C evolution is shown in panel (c). Relative capacity loss is higher in 50 °C cells (ca. 13.6 % on average) compared to 35 °C cells (ca. 7.7 % on average). Within the cells aged at same temperature, no systematic difference is observed between SOC levels. The observations on relative capacity change indicate that temperature is the dominant factor on SOH_C decrease of calendaric aging cells. Panel (d) shows SOH_C change as function of the square root of time

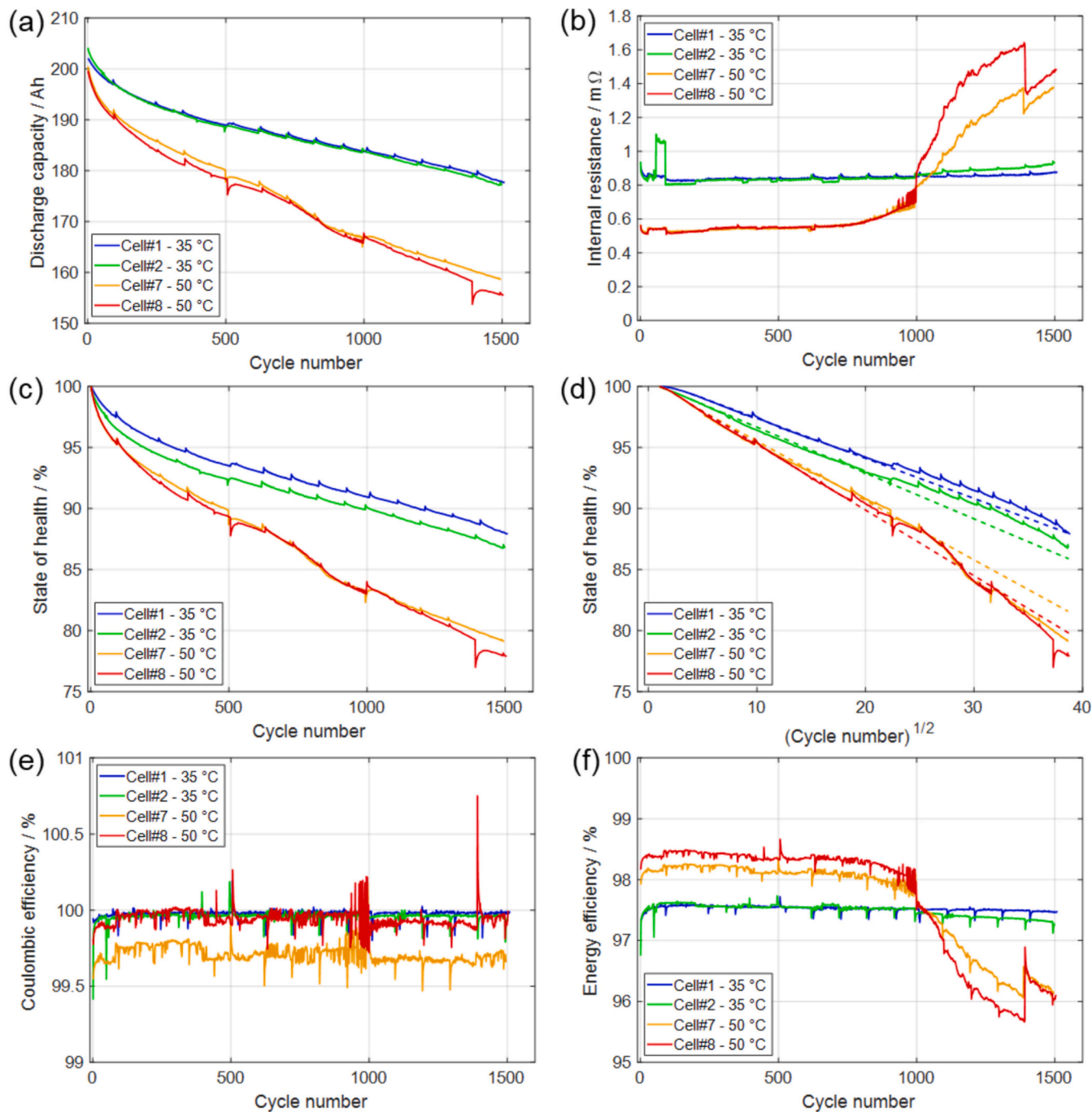


Fig. 6. Quantitative analysis of cycling data. Ambient temperature during cycling is given in figure next to cell numbers. (a) Discharge capacity, (b) internal resistance IR_{cycle} (Eq. (1)), (c) state of health SOH_C (Eq. (2)) vs. cycle number, (d) SOH_C vs. square root of cycle number, (e) coulombic efficiency CE (Eq. (3)), and (f) energy efficiency η_{energy} (Eq. (4)).

plotted along with dashed lines corresponding to linear fits. 50 °C cells, except for slight discrepancies at checkpoint #2, show a linear behavior with square root of time until the end of experiment. It is not possible to observe any linear behavior for 35 °C cells.

Panel (b) shows IR calculated from charge/discharge curves during checkups according to Eq. (1) (IR_{cycle}) as function of test time. After initial IR drop for all cells at checkpoint #1, all cells underwent IR increase at checkpoint #2. The highest IR values were measured for all cells except cell #9 (50 °C, 100 % SOC) at checkpoint #2. IR decreases once more at checkpoint #3 until rising finally close to initial values at the end of the

tests. IR measurements at checkpoint #3 show that the cells aged at 100 % SOC have lower IR than those aged at 75 % SOC, but this pattern could not be observed at other checkups. As discussed in Section 3.1, the IR measurements are subject to measurement uncertainties resulting from contact resistances that may vary from cell to cell and also from checkpoint to checkpoint due to the use of different battery cycler systems.

Panels (e) and (f) show CE and η_{energy} . The CE values are subject to limited measurement accuracy (cf. Section 2.3). Moreover, checkpoint #0 was performed with Biologic VMP3 battery cycler and checkups #1-4 with Basytec XCTS and Basytec GSM battery cyclers, with different

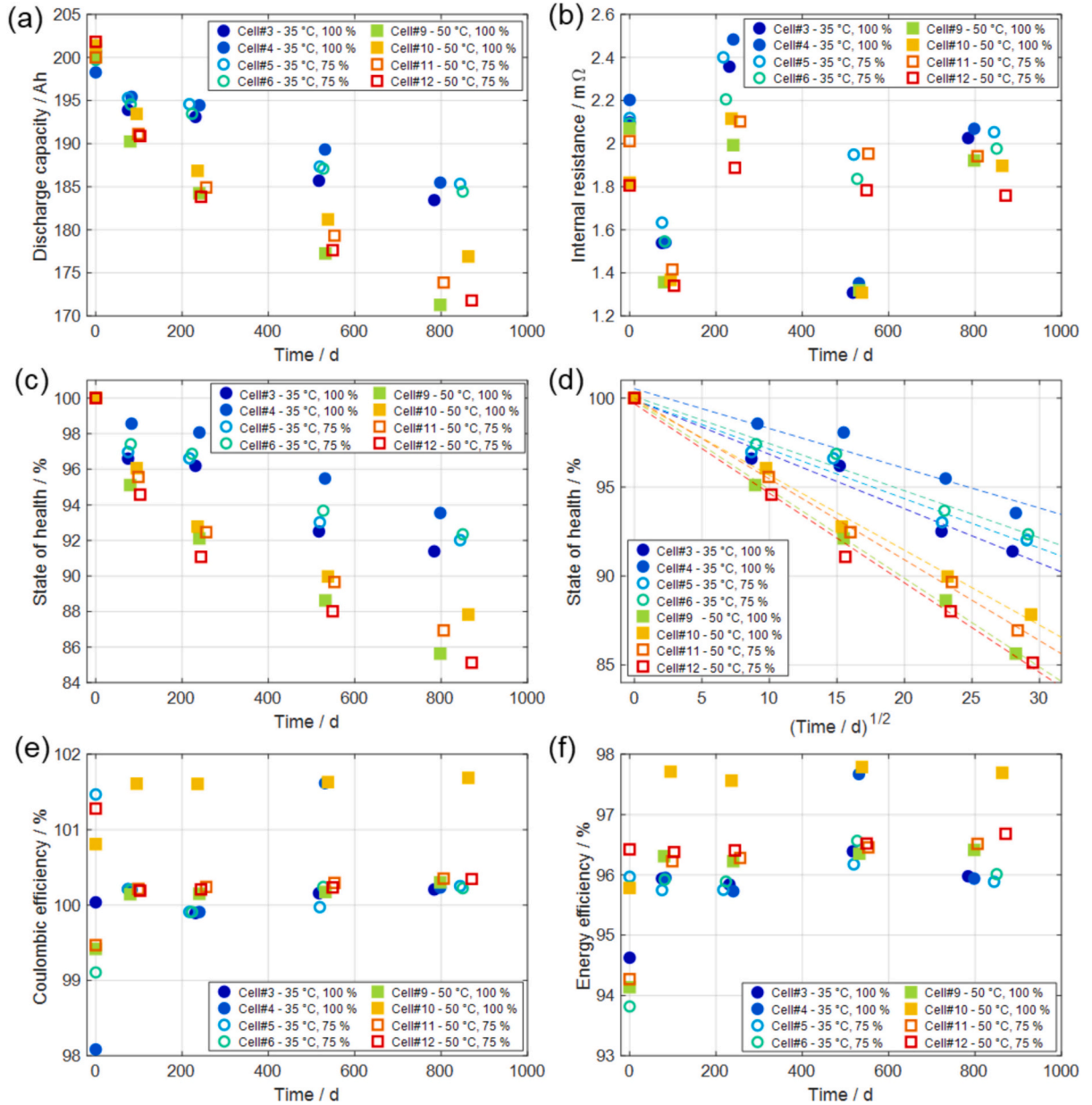


Fig. 7. Quantitative analysis of the checkup results of calendaric aging cells. Aging conditions (temperature and SOC) are given in legend next to cell numbers. Checkups were performed at 20 °C ambient temperature. (a) Discharge capacity, (b) internal resistance IR_{cycle} (Eq. (1)), (c) state of health SOH_c vs. time (Eq. (2)), (d) SOH_c vs. square root of time, (e) coulombic efficiency CE (Eq. (3)), (f) energy efficiency η_{energy} (Eq. (4)).

measurement accuracies. Qualitatively, for each individual cell, CE remains constant after checkup #1. The η_{energy} dataset in panel (f) shows that 50 °C cells have higher η_{energy} than 35 °C cells starting with checkup #1. Similar to CE , η_{energy} of calendaric aging cells show little variation (<0.5 % for all cells).

4. Degradation modes

4.1. Differential voltage analysis

Fig. 8 shows exemplary charge/discharge curves of cell #7 during cyclic aging at 50 °C along with the electrode balance analysis (experimental pOCV curve, synthetic OCV curve, DV plots of both experimental and model along with individual electrode signals) at four selected times corresponding to the start of tests (1st cycle), the midpoint of tests (750th cycle), last (1495th cycle), and the last checkup

(checkup #4). In panels (a) and (c), corresponding to the 1st and 750th cycle, well-formed voltage plateaus and steps are visible in the pOCV curves, which translate to clear peaks in the DV curves in panels (b) and (d). Synthetic OCV and DV curves match well with the experimental ones, so the optimization algorithm outputs are accurate. The DV peaks change their position towards lower charge throughput for the older cell; for example, the NE peak at ca. 65 Ah (cycle 1) shifts to ca. 45 Ah (cycle 750).

The data for cycle 1495 (panels e and f) show a qualitative difference. Here, the steps in the pOCV curve are hardly recognizable (panel e). Consequently, characteristic features of the DV curve (panel f) become very broad. The NE DV peak, which is initially positioned around 65 Ah, has nearly vanished, except for a small feature at ca. 25 Ah. As these peaks serve as a basis for identifying the lithium stoichiometry regions, the optimization algorithm is unable to obtain good results – the DV peak at ca. 25 Ah is shifted to ca. 5 Ah in the synthetic

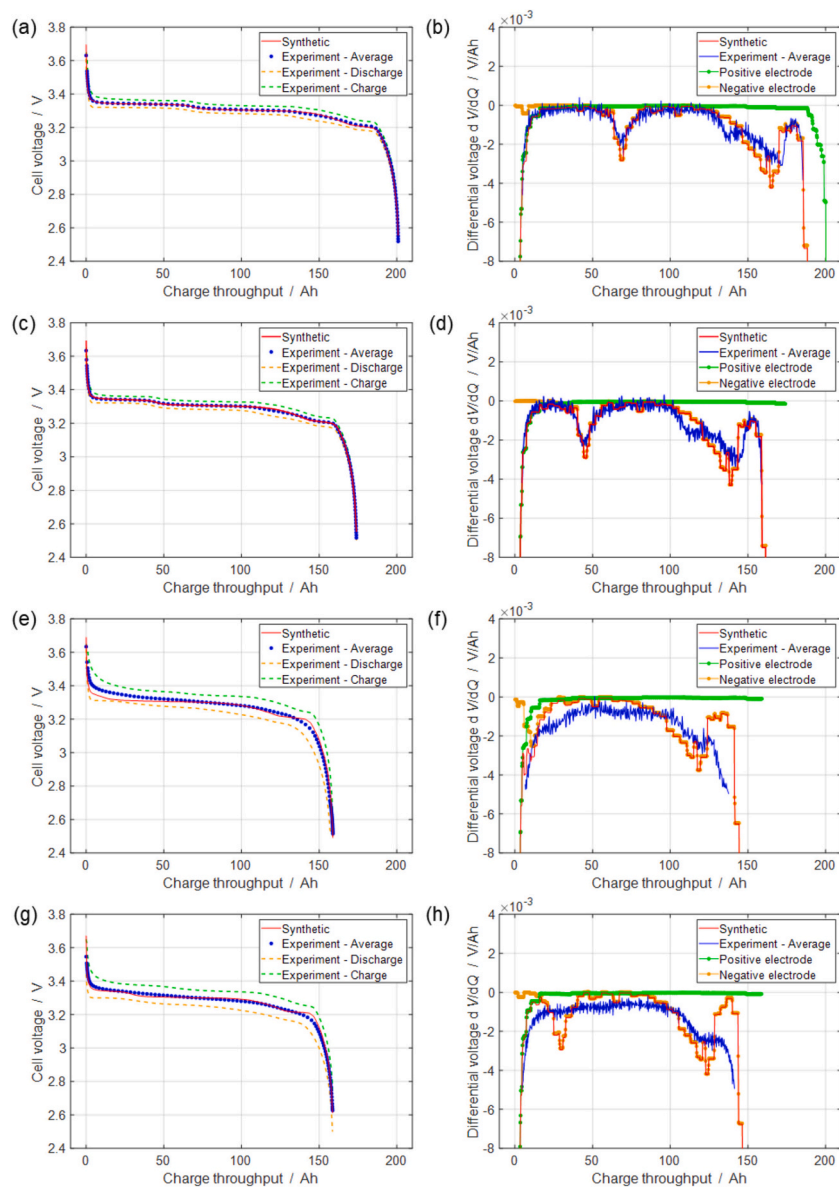


Fig. 8. Electrode balance analysis of cycling data and last checkup of cell #7. Left panels (a), (c), (e), and (g): Experimental charge, discharge, their average (pseudo-OCV), and synthetic OCV curve. Right panels (b), (d), (f), and (h): DV analysis of pseudo-OCV and synthetic OCV curves along with individual electrodes. (a–b) 1st cycle, (c–d) 750th cycle, (e–f) 1495th (last) cycle, (g–h) last checkup (checkup #4) performed at the end of cycling. Cycling tests were performed with the CCCV test protocol with 0.28C (0.05C CV cut-off current) at 50 °C, and the checkup #4 capacity test was performed with the CCCV test protocol with 0.1C (0.05C CV cut-off) at 20 °C.

curve, resulting in poor agreement between experimental and synthetic pOCV. The broadening of DV peaks can be attributed to spatially inhomogeneous aging, leading to a superposition of multiple different half-cell curves [28].

We have observed before that high C-rates broaden the characteristic steps in the voltage curves, as compared to low C-rates [35]. Therefore, low C-rates ($< C/10$) are used in literature for DV analyses [50]. The cyclic aging dataset in panels (a)–(f) consists of CCCV charge/discharge cycles at 0.28C at 50 °C. The checkup data in panels (g) and (h) was measured with CCCV 0.1C at 20 °C. Despite the lower C-rate, the DV peaks are broader in panel (h). Still, the optimization algorithm is able to converge to a good agreement between experimental and synthetic pOCV curves (panel g).

The broadening of DV peaks (as in panels f and h) beyond identifiable limits hinders the calculation of Li stoichiometries. The electrode balance analysis of cyclic aging cells showed that the peak identifiability of cells aging at 50 °C is lost after ca. 1000 cycles, corresponding to ca. 83

% SOH_C (cf. Fig. 6c). On the other hand, the broadening of peaks beyond identifiability does not occur for 35 °C cyclic aging cells (results not shown), allowing the calculation of the lithium stoichiometries until the end of the test. At the end of tests, the SOH_C of cells cycled at 35 °C are 86 % and 88 %.

4.2. Degradation modes of cyclic aging cells

The results of electrode balance and degradation mode analysis are shown for all cyclic aging cells in Fig. 9. Panels (a)–(d) show the minimum and maximum lithium stoichiometries of PE and NE as a function of cycle number. Two types of data sets are shown. The continuous data points (dots) represent the cycle-by-cycle analysis, the individual data points (circles) represent the analysis of pOCV curves performed during checkups (initial and after 500, 1000 and 1500 cycles).

The NE is almost completely delithiated in a fully-discharged cell

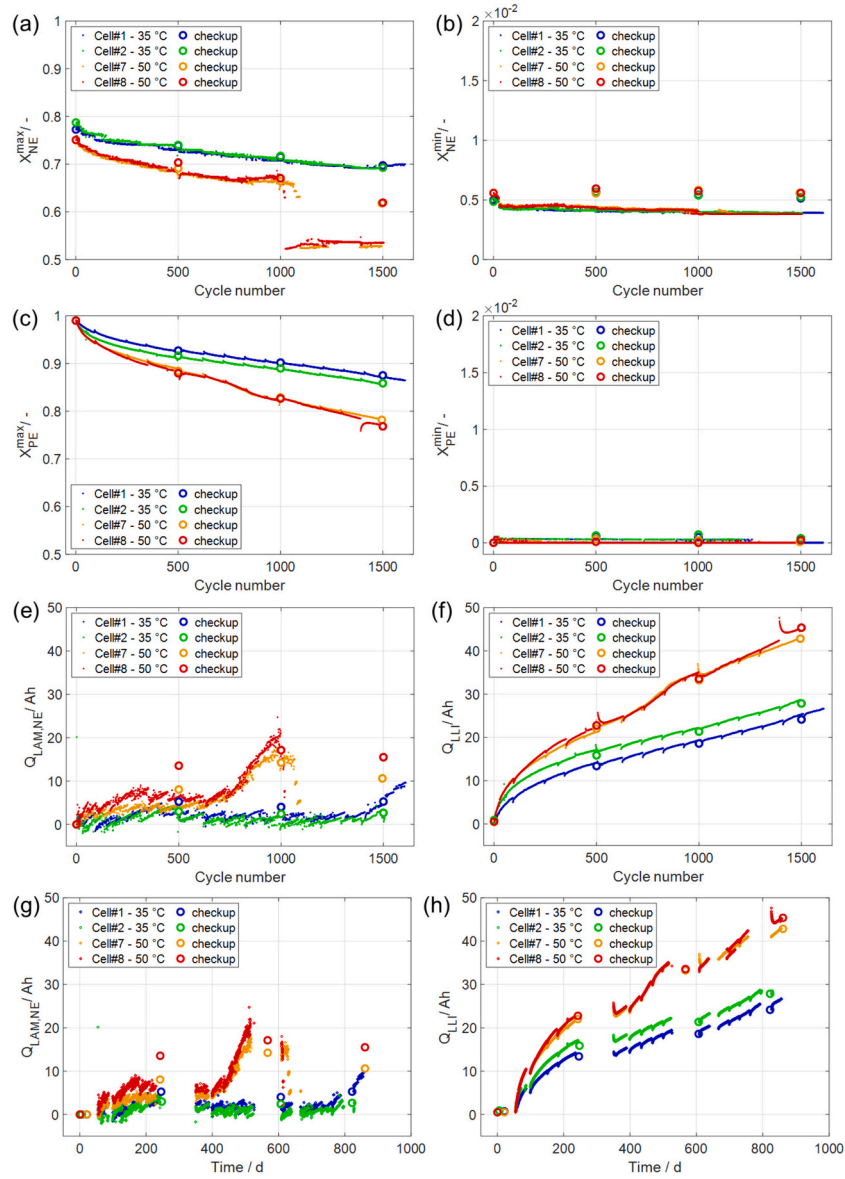


Fig. 9. Lithium stoichiometries and degradation modes resulting from electrode balance analysis of cyclic aging cells. Cyclic aging data is represented with dots (aging temperature is given next to cell number in legends), and checkup data (at 20 °C) is represented with circles. (a) Maximum NE stoichiometry, (b) minimum NE stoichiometry, (c) maximum PE stoichiometry, (d) minimum PE stoichiometry, (e) $Q_{LAM,NE}$ vs. cycle number, (f) Q_{LLI} vs. cycle number, (g) $Q_{LAM,NE}$ vs. time, (h) Q_{LLI} vs. time. All cycles were performed with the CCCV test protocol with 0.28C (0.05C CV cut-off current), and checkups were performed with the CCCV test protocol with 0.1C (0.05C CV cut-off).

(X_{NE}^{min} in panel b), and remains so during aging. Similarly, in a fully-charged cell, there are hardly any lithium ions left in PE (X_{PE}^{min} panel d). Thus, during the entire experiment, discharge cut-off voltage is limited by the NE, and charge cut-off voltage by the PE. PE maximum lithium stoichiometry (X_{PE}^{max} in panel c) starts at about 99 % lithiation for all cells. This shows that the usable cell capacity is initially dictated by PE. Upon aging, X_{PE}^{max} decreases proportional to the cell capacity according to Eq. (24). The maximum lithium stoichiometries of NE (X_{NE}^{max} in panel a) is below 80 % in the beginning of cycling; therefore, the NE is oversized by the manufacturer. Both X_{NE}^{max} and X_{PE}^{max} decrease during the entire aging process. In the case of the two 50 °C cells, a sharp drop of X_{NE}^{max} is observed after the 1000th cycle. As this does not seem realistic, we attribute it to the incorrect identifiability of the lithium stoichiometry due to broadened features (cf. previous section). The analyses resulting from the checkups (solid symbols) are consistent with those derived from the cycling data. This demonstrates the feasibility of electrode balance analysis from cycling data.

LAM at the NE, $Q_{LAM,NE}$, and LLI, Q_{LLI} , are shown in Fig. 9 both as a function of cycle number (panels e and f) and as a function of time (panels g and h). In the overall view, all cells exhibit both LLI and LAM_{NE} during aging, LLI is always higher than LAM_{NE}, and the losses are higher at 50 °C compared to 35 °C. The two degradation modes show quite different time behavior. LAM_{NE} initially slowly increases up to around 300 cycles, then shows a plateau behavior, and finally exhibits an “elbow”-like behavior of nonlinear increase. The “elbow” starts at ca. 1500 cycles at 35 °C, and at ca. 750 cycles at 50 °C. LLI, on the other hand, shows an initial fast and nonlinear increase up to ca. 300 cycles, after which the further loss increases almost linearly. It can be noted that LAM_{NE} data is missing after ca. 1000 cycles for the 50 °C cells; this is a consequence of the incorrect assessment of X_{NE}^{max} (cf. panel a) resulting from broadened DV peaks. The data resulting from the checkups (solid symbols) are consistent with the results derived from cycling only for LLI. For LAM_{NE}, the checkup results are not entirely consistent with the cycling results; the origin of the deviation is unclear.

4.3. Degradation modes of calendaric aging cells

For calendaric aging experiments, charge/discharge data is only available from the periodic checkups. Fig. 10 shows results of electrode balance and degradation mode analyses in a similar way as for the cyclic aging experiments in Fig. 9. Panels (a)–(d) show the NE and PE lithium stoichiometries variation. Maximum lithium stoichiometry values of both NE ($X_{NE}^{\max}$ in panel a) and PE ($X_{PE}^{\max}$ in panel c) continuously decrease for 50 °C cells and stay constant or decreases for 35 °C with aging. The decrease is always higher for 50 °C cells, regardless of SOC. Minimum NE stoichiometry ($X_{NE}^{\min}$ in panel b) and minimum PE stoichiometry

values ($X_{PE}^{\min}$ in panel d) remain at very small values. Similar to the cyclic aging cells, charge cut-off remains limited by the PE, and discharge cut-off by the NE.

The resulting degradation modes, LAM_{NE} and LLI , are shown in panels (e) and (f), respectively. LAM_{NE} values fluctuate irregularly around 10 Ah after an initial increase in checkup #1. There is no systematic difference between the two investigated SOC and temperature. LLI values exhibit a continuous increase for 50 °C cells with aging, regardless of storage SOC. For 35 °C, after the initial increase to ca. 10 Ah is observed in checkup #1, the values stay stagnant except for a small increase to ca. 12 Ah at the last checkup. The ambient temperature has a

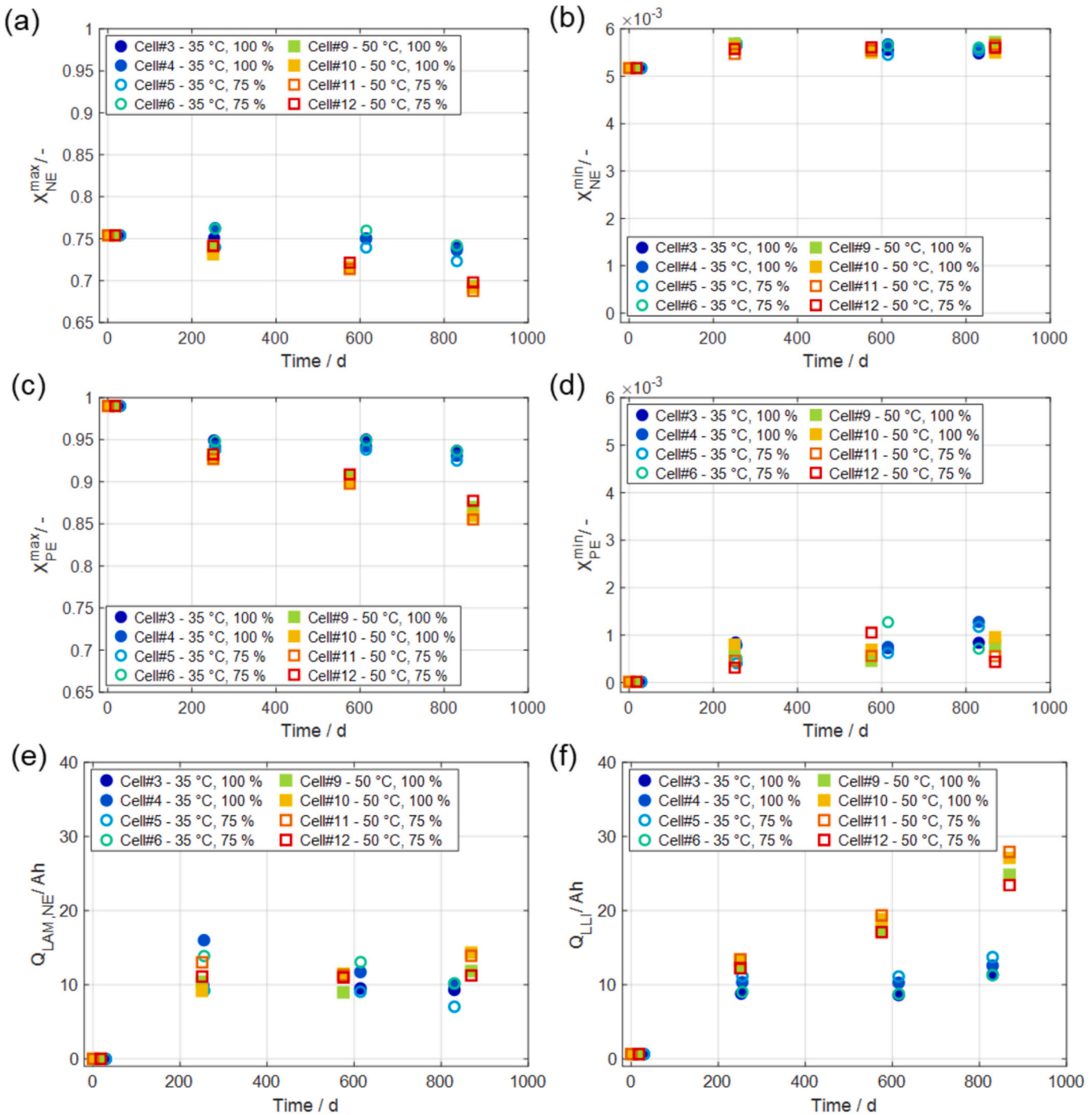


Fig. 10. Li stoichiometry and degradation modes calculated from electrode balance analysis of calendaric aging cells checkup measurements. (a) Maximum NE stoichiometry, (b) minimum NE stoichiometry, (c) maximum PE stoichiometry, (d) minimum PE stoichiometry, (e) $Q_{LAM,NE}$ vs. time, (f) Q_{LLI} vs. time. All checkups are performed using the CCCV test protocol with 0.1C (0.05C CV cut-off) at 20 °C.

significant effect on LLI, while the storage SOC (75 % vs. 100 %) does not have a systematic influence. Different to cyclic aging, LAM_{NE} and LLI are similar, at least at 35 °C.

5. Discussion

The objective of the present study is to investigate the influence of operating conditions (cycling vs. rest, temperature, and SOC) on the aging of large-format lithium-ion cells. In this section, the results presented in the previous sections are combined and interpreted. Fig. 11 shows SOH_C , IR, LAM_{NE} , and LLI of calendaric and cyclic aging cells in comparison. Only checkup results (measured at 20 °C for all cells) are presented in order to be able to compare cells aged under different conditions. In panels (c) and (d) the average LAM_{NE} and LLI values of two cells that underwent the same aging protocol (cf. Table 2) are shown for the sake of clearer comparison.

Fig. 11(a) compares SOH_C changes during checkups. The data show that, regardless of aging protocol, all cells aged at 50 °C had lower SOH_C than all cells aged at 35 °C at all checkups. This will be further discussed below. Fig. 11(b) compares the IR. Except for cycling aging at 50 °C, there is no clear trend of the IR. In particular, the storage SOC (75 % or 100 %) does not have a systematic influence. This is consistent with previous observations of calendaric aging of LFP cells by Keil et al. [51]. A systematic initial drop from checkup #0 to checkup #1 might be indicative of a conditioning behavior. As discussed in Section 3.1, because of the low IR of the studied large-format cells, the contact resistances can have a significant influence and can cause fluctuations of IR values. Note that, due to tests channel limitations, cell contacts had to be changed for some cells during each checkup for the calendaric aging

cells. Still, cyclic aging at 50 °C clearly shows a strong increase of IR at progressive aging.

As for the degradation modes, Fig. 11(c) compares LAM_{NE} . All aging conditions (calendaric and cyclic) show a strong increase after the initial checkup, after which $Q_{LAM,NE}$ remains on a more or less constant plateau throughout the remainder of the experiments. Aging conditions do not seem to systematically influence the initial increase, except for the 35 °C cyclic aging cells that clearly show the lowest increase. The physico-chemical origin of these observations (the initial increase, the consecutive plateau, the behavior of 35 °C cyclic cells) is unclear. Fig. 11(d) shows LLI. These data can be approximately divided into three aging groups: 1.) 50 °C cyclic aging cells, 2.) 50 °C calendaric aging cells and 35 °C cyclic aging cells, and 3.) 35 °C calendaric aging cells. The 50 °C cycling aging cells show continuously increasing Q_{LLI} values that are the highest at the end of tests (44.1 Ah in average). The 50 °C calendaric aging cells and 35 °C cyclic aging cells have continuously increasing and similar Q_{LLI} values at the end of experiments (in average 25.8 Ah and 26 Ah, respectively). The 35 °C calendaric aging cells have rather stagnant Q_{LLI} values throughout the measurements with a final value of 12.2 Ah in average. LLI is well correlated with SOH_C , indicating that LLI is the dominant mechanism for capacity loss.

For further comparison, we calculated the relative capacity loss ΔSOH_C at the end of experiments according to

$$\Delta SOH_C = 1 - SOH_{C, \text{checkup} \#4} \quad (25)$$

The values are presented in Table 3. Aging at 50 °C increased ΔSOH_C by factor of ca. two compared to aging at 35 °C, in average 16.5 % and 8.8 %, respectively. Among the cells that are aged at the same temperature, cyclic aging at 1C increased ΔSOH_C by factor of ca. 1.5 as

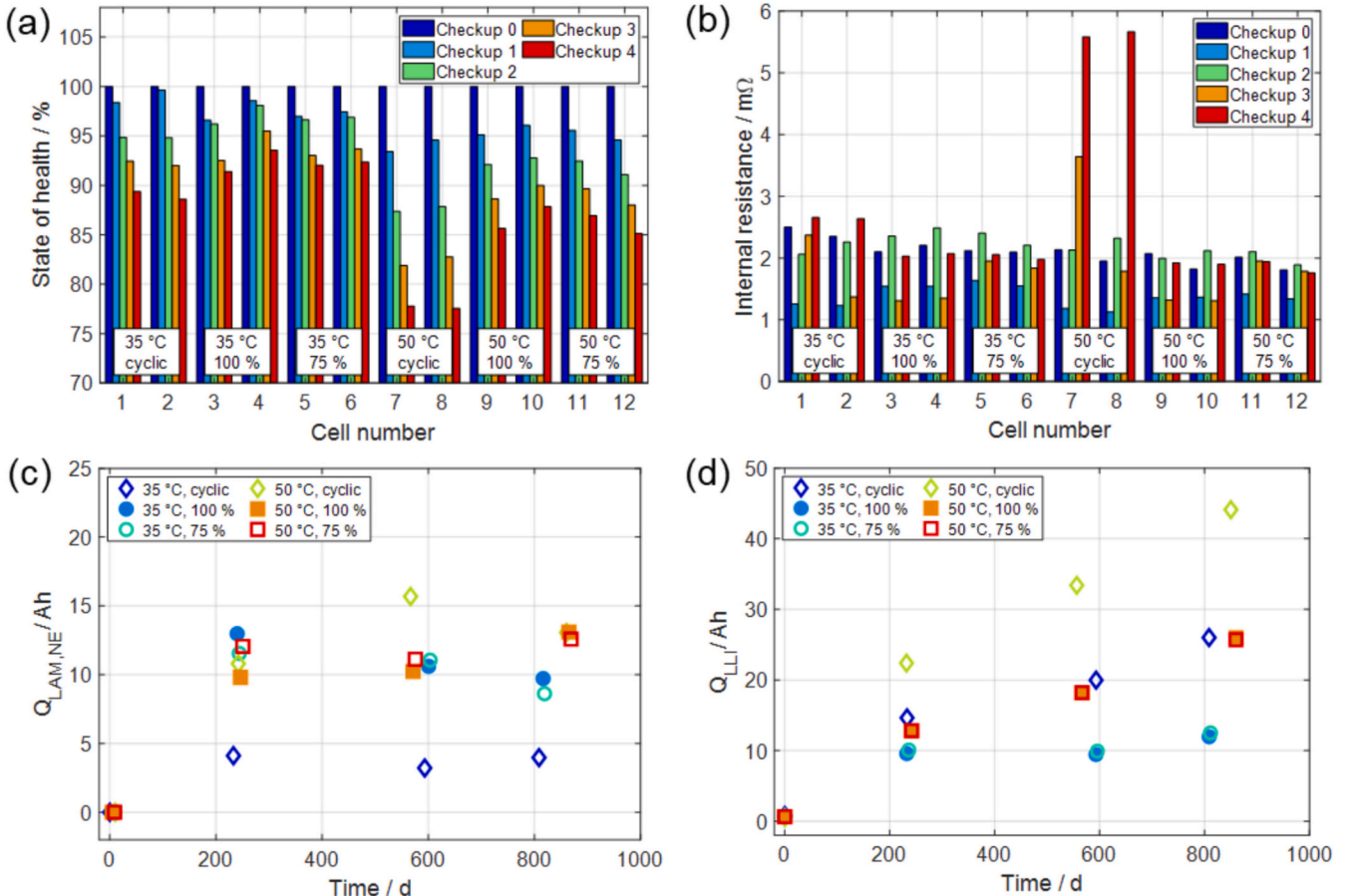


Fig. 11. Comparison of calendaric and cyclic aging investigations: (a) SOH_C vs cell number, (b) IR_{cycle} (Eq. (1)) vs. cell number, (c) average $Q_{LAM,NE}$ vs. time for each aging test protocol (d) average Q_{LLI} vs. time each aging test protocol.

Table 3Relative capacity loss ΔSOH_C for each cell at the end of experiment.

Temperature	35 °C						50 °C					
Aging protocol	Cyclic		Calendaric 75 % SOC		Calendaric 100 % SOC		Cyclic		Calendaric 75 % SOC		Calendaric 100 % SOC	
Cell number	1	2	3	4	5	6	7	8	9	10	11	12
ΔSOH_c (Eq. (25))	10.6	11.4	8.6	6.4	8.0	7.6	22.2	22.4	14.4	12.2	13.1	14.9
Average ΔSOH_c by aging protocol	11.0		7.5		7.8		22.3		13.3		14.0	
Average ΔSOH_c by temperature	8.8						16.5					

Table 4Relative loss ΔSOH_C during cyclic and calendaric aging as well as their ratio for this work compared to LFP/graphite cell aging studies by Schimpe et al. [56] and Safari et al. [14].

	Cyclic capacity loss at 1C / %	Calendaric capacity loss at 100 % SOC / %	Ratio
This work, 35 °C, 28 months	11	7.8	1.41
This work, 50 °C, 28 months	22.3	14	1.59
Schimpe et al., 35 °C, 8 months	9.2	5.8	1.59
Schimpe et al., 55 °C, 2 months	10	5.2	1.92
Safari et al., 25 °C, 6 months	6.3	3.4	1.85
Safari et al., 45 °C, 6 months	15.4	9.4	1.64

compared to calendaric aging at 100 % SOC. Table 4 shows that somewhat higher ratios were observed in other studies. Additionally, for calendaric aging at both temperatures, no distinctive difference of ΔSOH_C is observed between 75 % and 100 % SOC (the difference is 0.3 % for 35 °C, and 0.7 % for 50 °C cells). The two SOC levels were selected (cf. Section 2.2) because they cause a significant difference in PE potential while having the same NE potential. The results show that PE potential has either little influence on PE aging, or that the influence is masked by the more dominant LLI. The influence of NE potential on calendaric aging was not subject of the present study; it was shown before to have a strong impact on capacity loss [52].

We have further investigated the influence of temperature on capacity loss by an Arrhenius analysis for all investigated cells combined. When comparing cyclic and calendaric aging cells, it needs to be kept in mind that cycling causes heat losses and therefore increases internal cell temperature. Initial thermal simulations using a pseudo-3D model [53] parameterized to the present large-format prismatic cell indicated an average temperature inside the cell of ca. 3 °C above the ambient temperature for 50 A cycling. This increased temperature can further accelerate aging. Assuming an average temperature 3 °C above ambient temperature for the cycling cells, the resulting Arrhenius plot is shown in Fig. 12. While the twelve cells do not fall entirely together on a single Arrhenius slope (the cyclic aging cells are above the slope, the calendaric

aging cells below), the data still demonstrate that temperature has a strong influence on aging independently of aging protocol. The obtained global activation energy is (37.3 ± 12.8) kJ/mol. This represents an average value over all cells and aging protocols. The rather large value of 12.8 kJ/mol of the 95 % confidence bound is indicative of the deviation from ideal Arrhenius behavior. Still, Fig. 12 shows that temperature dominates other aging drivers in the investigated cells. This is consistent with the conclusion of our previous work on similar cells operated in battery systems [9].

6. Summary and conclusions

The influence of aging protocol (cyclic or calendaric), temperature (35 °C or 50 °C), and storage SOC (75 % or 100 %) on cell aging was investigated by testing twelve large-format LFP-based lithium-ion cells under controlled laboratory conditions. Cyclic aging was investigated for 1500 cycles, and calendaric aging for 850 days. Both the cycling data, and data resulting from periodic reference performance tests, were investigated by degradation mode analysis in order to quantify LAM and LLI. The main results are summarized as the following:

- Regardless of aging conditions, all cells exhibited significant capacity losses over time. Within the duration of the aging experiments, an increase of IR was only observed late during 50 °C cyclic aging, and not at all for calendaric aging. Therefore, capacity fade was the primary aging indicator for both calendaric and cyclic aging under the investigated conditions.
- Cyclic aging cells tested at 35 °C lost, on average, 11 % of their initial capacity, whereas the cyclic aging cells tested at 50 °C lost, on average, 22.3 % of their initial capacity at the end of the experiment. 50 °C cyclic aging cells reached the “end of life” (20 % capacity loss) at ca. 1350 cycles. Therefore, capacity retention during cyclic aging was superior at 35 °C.
- All calendaric aging cells tested at 50 °C lost higher capacity (ca. 14 % on average) than calendaric aging cells tested at 35 °C (ca. 8 % on average), independent of storage SOC level. No distinctive effect of the investigated two storage SOC levels (75 % and 100 %) on aging was observed. This shows that the LFP electrode potential has no influence on observed capacity loss.
- Temperature was observed to be the dominant aging driver (rather than aging protocol or storage SOC). For example, calendaric aging at 50 °C caused faster aging than cyclic aging at 35 °C. The observed global activation energy for capacity loss (over all conditions) was 37.3 kJ/mol.

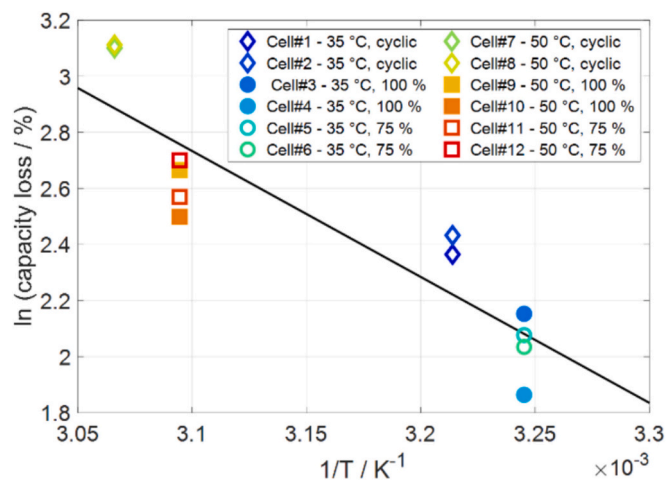


Fig. 12. Arrhenius plot of the capacity losses of all cells at the end of experiment.

- The energy efficiency at 35 °C cyclic aging was constantly above 97 %. In the first 1000 cycles, the energy efficiency at 50 °C cyclic aging was above 98 %, but after the efficiency dropped to 96 % as IR increased. Energy efficiency remained nearly constant during calendar aging, with minor fluctuations between 96 and 98 %. Coulombic efficiency was unity (within the measurement accuracy) during cyclic and calendar aging.
- Degradation modes could be successfully obtained from the cycling data, without requiring low-C-rate pOCV tests. LLI was observed the dominating degradation mode for cyclic aging, exhibiting a significant temperature dependence. For calendar aging, LAM_{NE} and LLI were observed to have a similar magnitude. LAM_{NE} did not exhibit a clear dependence on temperature or cycling type. LAM_{PE} could not be quantified due to the plateau in the LFP half-cell voltage curve.
- A square-root-of-time dependence of LLI was observed until ca. 10 % capacity loss for all cyclic aging cells, indicating a diffusion-limited SEI growth mechanism. However, at later stages of aging (SOH_C < 90 %), LLI cannot be explained by a single $t^{0.5}$ -dependent mechanism.

In conclusion, the results show the dominant effect of temperature on aging of LFP cells. The effect of aging protocol (cyclic or calendar aging) has secondary influence. The investigated commercial cells are intended to be used for stationary storage in a residential BESS. Therefore, the results suggest that a good thermal management and, potentially, active cooling are important extending the lifetime and therefore improving the economic viability of BESS.

CRediT authorship contribution statement

Mehmet C. Yagci: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Oliver Richter:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **René Behmann:** Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Wolfgang G. Bessler:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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